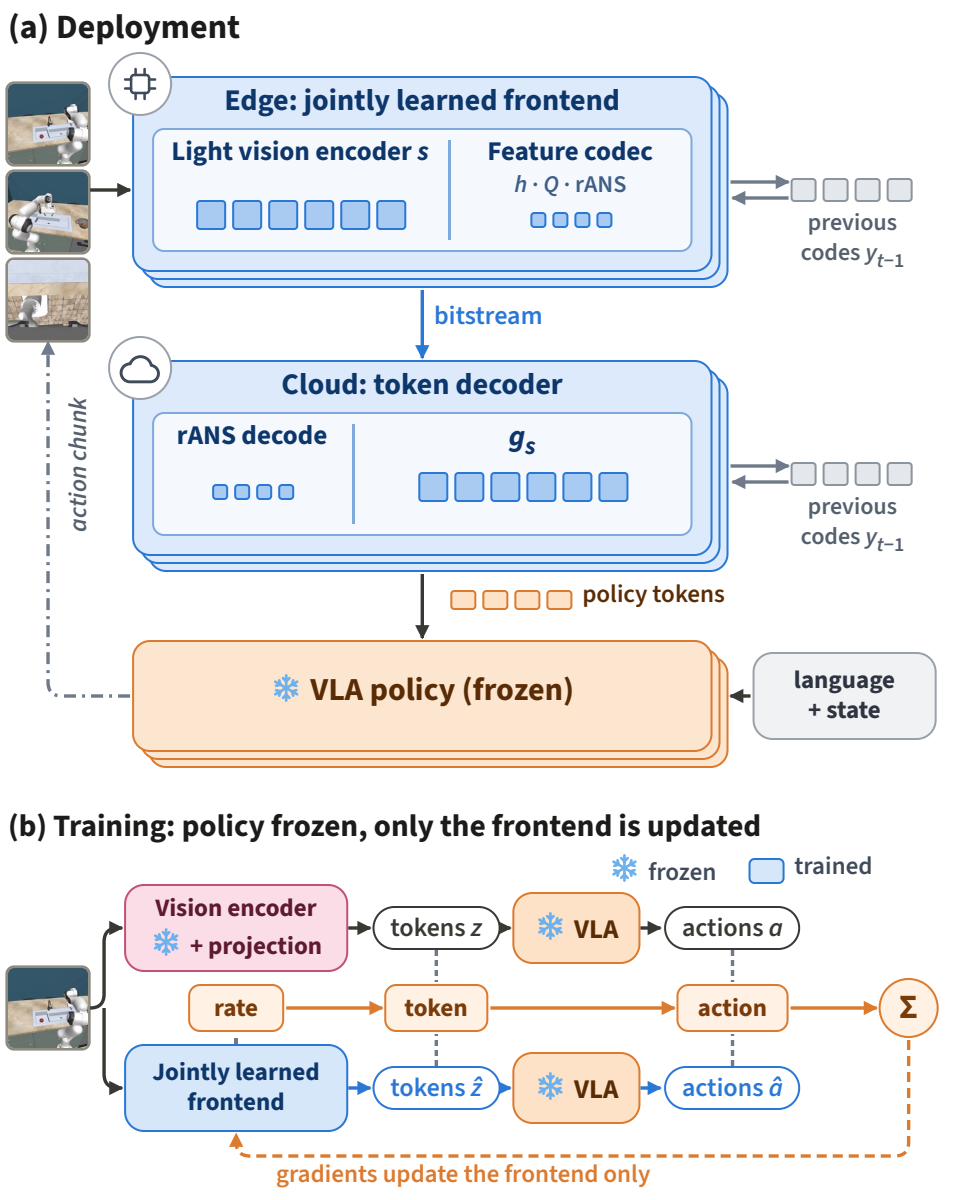

(a) Deployment
Edge: jointly learned frontend
Light vision encoder s
Feature codec
h · Q · rANS
previous codes y_{t-1}
bitstream
Cloud: token decoder
rANS decode
g_s
previous codes y_{t-1}
action chunk
policy tokens
VLA policy (frozen)
language + state
(b) Training: policy frozen, only the frontend is updated
frozen
trained
Vision encoder + projection
tokens z
VLA
actions a
rate
token
action
Σ
Jointly learned frontend
tokens ẑ
VLA
actions â
gradients update the frontend only